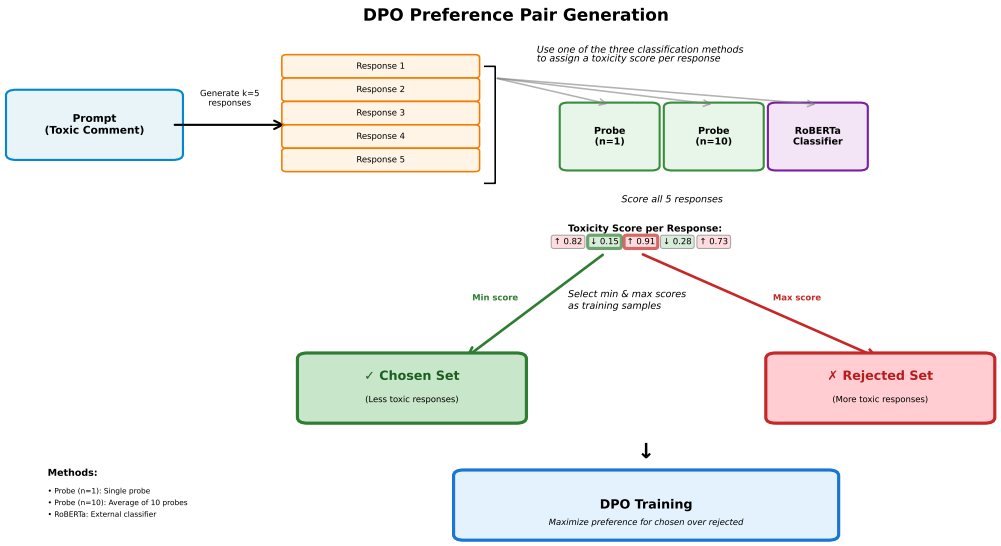


Figure 1: Probe-based DPO method

where π_θ is the policy being optimized, π_{ref} is the frozen reference model (the base model), y^+ and y^- denote the chosen and rejected responses respectively, $\beta = 0.1$ is the KL penalty coefficient, and σ is the sigmoid function. We use LoRA Hu et al. [2022] for parameter-efficient fine-tuning.

3.2.3 Experimental Conditions

We evaluate three DPO conditions:

- **DPO-Classifier:** Standard DPO using preference pairs as judged by the RoBERTa toxicity classifier. This serves as our baseline and does not involve probes.
- **Single DPO-Probe:** DPO using preference pairs as judged by a single trained probe.
- **10 DPO-Probes:** DPO using preference pairs as judged by combining predictions from 10 independently trained probes (averaged probabilities).

We don't expect probe-based DPO to outperform output-based DPO for training against toxicity, since toxicity can be well detected from the outputs. Our experiments using toxicity should be taken more as a proof of concept that probe-based training can successfully reduce toxicity. Further research should evaluate the success of this training scheme in preventing deception and dishonesty.

3.3 Supervised Fine-Tuning with Probe-Based Regularization

We introduce a probe-based regularization approach for supervised fine-tuning (SFT) that penalizes the model for generating representations that the probe classifies as toxic. This method directly incorporates toxicity detection into the training objective, steering the model away from toxic generation patterns at the representation level.

This could be useful if one wants to train on a dataset that contains toxic content (eg an unfiltered scrape from social media) and wants to learn the information in it without making the model more toxic.

3.3.1 Training Objective

The standard language modeling objective for SFT is the negative log-likelihood:

$$\mathcal{L}_{\text{LM}} = -\mathbb{E}_{x\sim\mathcal{D}}\left[\log p_\theta(x)\right] \tag{2}$$

where x is a training sequence and θ represents the model parameters.

We augment this objective with a probe-based penalty term. During training, we extract the hidden state activations $h \in \mathbb{R}^d$ from the selected layer (layer 20) and compute the mean-pooled activation across all tokens in each training batch. The probe then predicts a toxicity probability $p_{\text{probe}}(h)$ for these activations. Our combined training objective becomes:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{LM}} + \lambda \cdot p_{\text{probe}}(h) \tag{3}$$

where λ is a hyperparameter controlling the strength of the probe-based regularization. Importantly, the probe parameters are frozen during SFT—only the language model parameters θ are updated. This ensures gradients flow through the probe to update the model's representations without modifying the probe itself.

We fine-tune the model using LoRA Hu et al. [2022] for 1 epoch on 10,000 samples from the Civil Comments training set.

3.3.2 Multi-Probe Aggregation

When training with multiple probes, there are different ways of aggregating their scores into a single score. We experiment with this in Appendix B. Our experiments find that it's best to use one probe at a time and cycle them sequentially through training batches. This leverages the diversity of multiple independently trained probes while maintaining deterministic training dynamics. In the same experiment, we also evaluate how much priority to give to the probe-loss and find $\lambda = 1$ to be a good value.

4 Experiment

4.1 Dataset and Task

We conduct our experiments using the Civil Comments dataset Duchene et al. [2023], a large-scale toxicity detection corpus containing approximately 2 million public comments from online discussions. We focus exclusively on comments labeled as toxic (toxicity score ≥ 0.5) to create a challenging training environment where the model must learn to respond non-toxically to inherently toxic prompts. This design choice tests whether our probe-based methods can effectively steer model behavior when confronted with adversarial inputs that might naturally elicit toxic responses.

Each comment is formatted into a conversational prompt using the template: "Respond to the following user comment: [comment text]". This framing simulates a realistic scenario where language models must handle toxic user inputs while maintaining appropriate responses.

We use the Gemma-3-1B model Team et al. [2025], which has not had alignment training applied. This is important for the experiments, as the instruction-tuned model barely shows toxic responses.

4.2 Evaluation Metrics

We track multiple metrics to assess both toxicity reduction and potential side effects:

Toxicity Rate: The fraction of model generations classified as toxic (score ≥ 0.5) by the RoBERTa toxicity classifier on 100 held-out toxic prompts. Lower values indicate more successful detoxification.

Probe AUC - Training Probes: Area under the ROC curve for the probes used during training, evaluated on post-training activations. This measures whether the model's internal representations remain detectable as toxic according to the original training signal.

Probe AUC - Held-Out Probes: AUC for 2 independently trained probes (trained on separate data subsets) never exposed during model training. High AUC here indicates the model hasn't simply learned to fool the training probes but has genuinely altered its representations.

Probe AUC - Retrained Probes: AUC for 2 probes retrained from scratch on post-training activations from held-out data. This tests whether toxicity remains linearly detectable in the model's representation space after training. Low AUC suggests the model has fundamentally reorganized how it represents toxic content.

Perplexity on the training distribution: Measured on 1,000 held-out toxic comments from Civil Comments. This assesses whether probe-based training impedes the model's ability to fit the training distribution.

Perplexity on Wiki: Measured on 1,000 samples from WikiText-2. This assesses whether probe-based training degrades general language modeling capabilities on out-of-distribution clean text.

5 Results

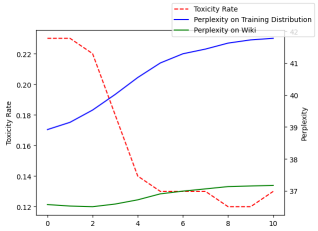
This section shows the experimental results. It should be noted that results are noisy and were only run on one seed, thus limiting the strength of conclusions we can draw.

5.1 DPO

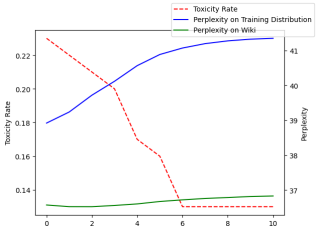
We present results from three DPO conditions: classifier-based (baseline), single probe-based, and 10 probe-based. Table 1 shows final metrics, while Figure 2 displays training dynamics.

	Toxicity Rate	AUC train probes	AUC held-out probes	AUC retrained probes	Perplexity Training Distribution	Perplexity Wiki
Classifier	0.11	-	0.866	0.770	41.78	37.18
1 probe	0.14	0.977	0.938	0.926	41.35	37.03
10 probes	0.15	0.987	0.994	0.992	41.42	36.91

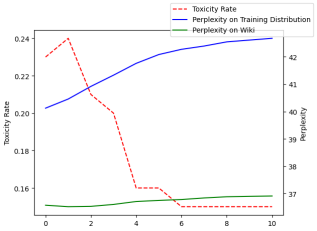
Table 1: Final results of DPO training.



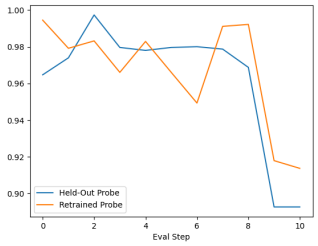
(a) Toxicity rate, perplexity on training distribution, and perplexity on the Wiki-dataset during DPO training using a **text-based toxicity classifier**.



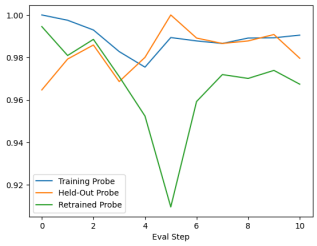
(b) Toxicity rate, perplexity on training distribution, and perplexity on the Wiki-dataset during DPO training using **one probe-based classifier**.



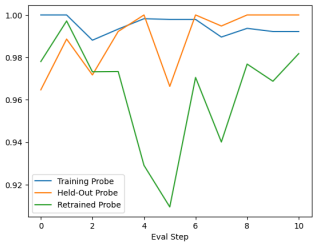
(c) Toxicity rate, perplexity on training distribution, and perplexity on the Wiki-dataset during DPO training using **10 probe-based classifiers**.



(d) Area Under Curve (AUC) of held-out and newly trained probes during DPO training using a **text-based toxicity classifier**.



(e) Area Under Curve (AUC) of the probe used for training, held-out, and newly trained probes during DPO training using **one probe-based classifier**.



(f) Area Under Curve (AUC) of probes used for training, held-out, and newly trained probes during DPO training using **10 probe-based classifiers**.

Figure 2: Changes in probe accuracy, toxicity rate, and model capability when training with DPO using a text-based classifier, one probe-based classifier, or 10 probe-based classifiers.